Correlation and Covariance

Q1. Define Covariance and explain how it differs from Correlation in terms of scale and interpretation.

Answer :-

- Covariance measures how two variables change together. It is defined as:

$$\text{Cov}(X,Y) = \frac{1}{n}\sum(X_i - \bar{X})(Y_i - \bar{Y})$$

  - If both variables increase together → positive covariance
  - If one increases while the other decreases → negative covariance

- **Key Difference from Correlation:**

Aspect	Covariance	Correlation
Scale	Depends on units of X and Y	Unit-free (standardized)
Range	No fixed range (can be any value)	Always between -1 and +1
Interpretation	Hard to interpret magnitude	Easy to interpret strength

- Correlation is essentially a *normalized version* of covariance:

$$\boldsymbol{r = \frac{\text{Cov}(X,Y)}{\sigma_X \sigma_Y}}$$

Q2. What does a positive, negative, and zero covariance indicate about the relationship between two variables?

Answer :-

- Positive covariance (> 0):
  - Variables move in the **same direction**
  - (e.g., height and weight)
- Negative covariance (< 0):
  - Variables move in **opposite directions**
  - (e.g., price and demand)
- Zero covariance (= 0):
  - **No linear relationship**
  - (but a nonlinear relationship might still exist)

Q3. Discuss the limitations of covariance as a measure of relationship between two variables. Why is correlation preferred in many cases?

Answer :-

Limitations:

1. **Scale-dependent** → changes with units (e.g., meters vs cm)
2. **Hard to interpret magnitude** → no standard range

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3. **Not comparable across datasets**

4. Only captures **linear relationships**

Why correlation is preferred:

- Standardized (range -1 to +1)
- Easy to interpret strength and direction
- Comparable across different datasets

Q4. Explain the difference between Pearson's correlation coefficient and Spearman's rank correlation coefficient. When would you prefer to use Spearman's correlation?

Answer :-

- **Pearson vs Spearman Correlation**

Feature	Pearson Correlation	Spearman Rank Correlation
Measures	Linear relationship	Monotonic relationship
Data type	Continuous, normally distributed	Ordinal or non-normal data
Sensitive to outliers	Yes	Less sensitive
Based on	Actual values	Ranks

When to use Spearman:

- Data is **not normally distributed**
- **Relationship is nonlinear but monotonic**
- **Presence of outliers**
- **Data is ordinal (ranked)**

Q5. If the correlation coefficient between two variables X and Y is 0.85, interpret this value in context. Can you infer causation from this value? Why or why not?

Answer :-

- Interpretation of Correlation = 0.85
  - A value of **0.85** indicates:
    - Strong positive relationship
    - As X increases, Y tends to increase significantly

Can we infer causation?

❌ **No**

Why not?

- Correlation only shows association, not cause-effect

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- There could be:

  - A **third variable (confounder)**
  - **Reverse causation**
  - **Coincidence**

- Example **:**

  Ice cream sales and drowning incidents may be correlated due to hot weather—not because one causes the other.

Q6. Using the dataset below, calculate the covariance between X and Y.

X	2	4	6	8
Y	3	7	5	10

Answer :-

X = 2, 4, 6, 8

Y = 3, 7, 5, 10

Mean values:

$\bar{X} = 5$, $\bar{Y} = 6.25$

Using the covariance formula:

$$Cov(X,Y) = \frac{1}{n}\sum(Xi - \bar{X})(Yi - \bar{Y})$$

X	Y	X - $\bar{X}$	Y - $\bar{Y}$	Product
2	3	-3	-3.25	9.75
4	7	-1	0.75	-0.75
6	5	1	-1.25	-1.25
8	10	3	3.75	11.25
20	25	-	-	19

Sum = 19 , n = 4

$$Cov(X,Y) = \frac{1}{n}\sum(Xi - \bar{X})(Yi - \bar{Y})$$

$$= \frac{19}{4}$$

$$= 4.75$$

Answer: Covariance = 4.75

Q7. Compute the Pearson correlation coefficient between variables A and B:

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A	10	20	30	40	50
B	8	14	18	24	28

Answer :-

A = 10, 20, 30, 40, 50

B = 8, 14, 18, 24, 28

Mean values:

$A^- = 30$, $B^- = 18.4$

A	B	A - A-	B - B-	Product	$(A - 30)^2$	$(B - 18.4)^2$
10	8	-20	-10.4	208	400	108.16
20	14	-10	-4.4	44	100	19.36
30	18	0	-0.4	0	0	0.16
40	24	10	5.6	56	100	31.36
50	28	20	9.6	192	400	92.16
150	**92**			**500**	**1000**	**251.2**

Formula:

$$r = \frac{\sum(x-x^-)(y-y^-)}{\sqrt{\sum(A-\text{A}-)2*\sum((B-\text{B}-)2}}$$

$$= \frac{500}{\sqrt{1000*251.2}}$$

$$= \frac{500}{\sqrt{251200}}$$

$$= \frac{500}{501.2}$$

r = 0.998

Observations:-

- As A increases, B increases steadily → strong positive linear trend.
- Calculating gives:

Answer: $r \approx 0.995$(very strong positive correlation)

Q8. The following table shows heights (in cm) and weights (in kg) of 5 students. Find the correlation coefficient between Height and Weight.

Height (X)	150	160	165	170	180
Weight(Y)	50	55	58	62	70

Answer :-

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Mean values:

$\bar{X} = 165$, $\bar{Y} = 59$ n = 5 $(\mathbf{X} - \bar{\mathbf{X}})^2$

Formula:

$$r = \frac{\sum(x-\bar{x})(y-\bar{y})}{\sqrt{\sum(A-\mathrm{A}-)2*\sum((B-\mathrm{B}-)2}}$$

$$= \frac{335}{\sqrt{500*228}}$$

$$= \frac{335}{\sqrt{114000}}$$

$$= \frac{335}{337.6}$$

r = 0.992

Conclusion: Very strong **positive correlation** (height badhane par weight bhi badh raha hai)

Q9. Given the dataset below, determine whether there is a positive or negative correlation between X and Y. (No need for exact calculation, just reasoning.)

X	1	2	3	4	5
Y	15	12	9	7	3

Answer :-

Negative correlation

Observation:

- X increase ho raha hai (1 → 5)
- Y decrease ho raha hai (15 → 3)

Conclusion:

- Jab ek variable badhta hai aur doosra ghatta hai, toh Negative Correlation hota hai

Q10. Two investment portfolios have the following returns (%) over 5 years. Compute the covariance and correlation coefficient, and interpret whether the portfolios move together.

Year	Portfolio A	Portfolio B
1	8	6
2	10	9
3	12	11
4	9	8
5	11	10

Answer :-

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- If this is asking for Mean (Average):-

Mean = $\frac{5+11+10}{3}$

= $\frac{26}{3}$

= 8.67

- If this is asking for Median:-

Arrange: 5, 10, 11

Middle value = **10**

- If this is asking for Mode:-

No repeating value → **No mode**

- If this is part of Correlation / Covariance**:-**

Then this row alone is incomplete (we need another row like X & Y).